

CodeAlpha C++ Internship

TASK 1: CGPA Calculator

Complete source code for this internship task.

```
#include <iostream>
#include <vector>
#include <iomanip>
using namespace std;

int main() {
    int courses;
    cout << "Enter number of courses: ";
    cin >> courses;

    double totalCredits = 0, totalPoints = 0;
    vector<double> grades(courses), credits(courses);

    for (int i = 0; i < courses; i++) {
        cout << "\nCourse " << i + 1 << " grade point: ";
        cin >> grades[i];
        cout << "Course " << i + 1 << " credit hours: ";
        cin >> credits[i];

        totalCredits += credits[i];
        totalPoints += grades[i] * credits[i];
    }

    double gpa = (totalCredits > 0) ? totalPoints / totalCredits : 0;

    cout << fixed << setprecision(2);
    cout << "\nSemester GPA = " << gpa << endl;
    cout << "Final CGPA = " << gpa << endl;

    return 0;
}
```